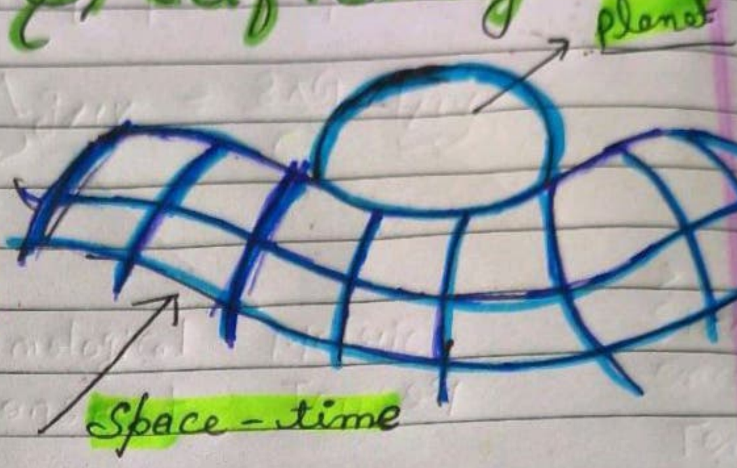
What Specifically Bends Space-time



The Sources of space time Curvature are organized into single object the stress-energy tensor Consists of

1. Mass-Energy Density
2. Pressure & stress
3. Momentum & flow
4. Energy of fields

Internal pressure & mechanical pressure also deforms space-time

Any object with mass & Energy deforms Space & time

The movement of mass and Energy cause Curvature

Empty space containing electromagnetic fields that posses energy also bends Curvature

Mechanism of Bending

- Curvature of Time

What we perceives as gravity is actually the warping of time rather than space.

- Intrinsic Curvature

Spacetime does not necessarily had to bend into another dimension but instead, its internal geometry changes.

- Geodesics: In a bent spacetime objects follows the "Straightest path possible" therefore a Curved looking orbit to us is actually a straight path through a 4-D geometry.

Einstein's Field Equation

$$\underbrace{G_{\mu\nu}}_{\text{Einstein Tensor}} + \underbrace{\Lambda}_{\text{Cosmological Constant}} \underbrace{g_{\mu\nu}}_{\text{Metric Tensor}} = \frac{8\pi G}{c^4} \underbrace{T_{\mu\nu}}_{\text{Stress energy Tensor}}$$

$C \rightarrow$ Speed of light

$G \rightarrow$ Newton's gravitational constant

$G_{\mu\nu}$ describes the curvature of space-time

$g_{\mu\nu}$ describes the geometry of space time

$T_{\mu\nu}$ describes the stress energy momentum & mass distribution.

Associated Phenomena

- Gravitational Lensing
- Black holes
- Gravitational waves
- Cosmologic RedShift